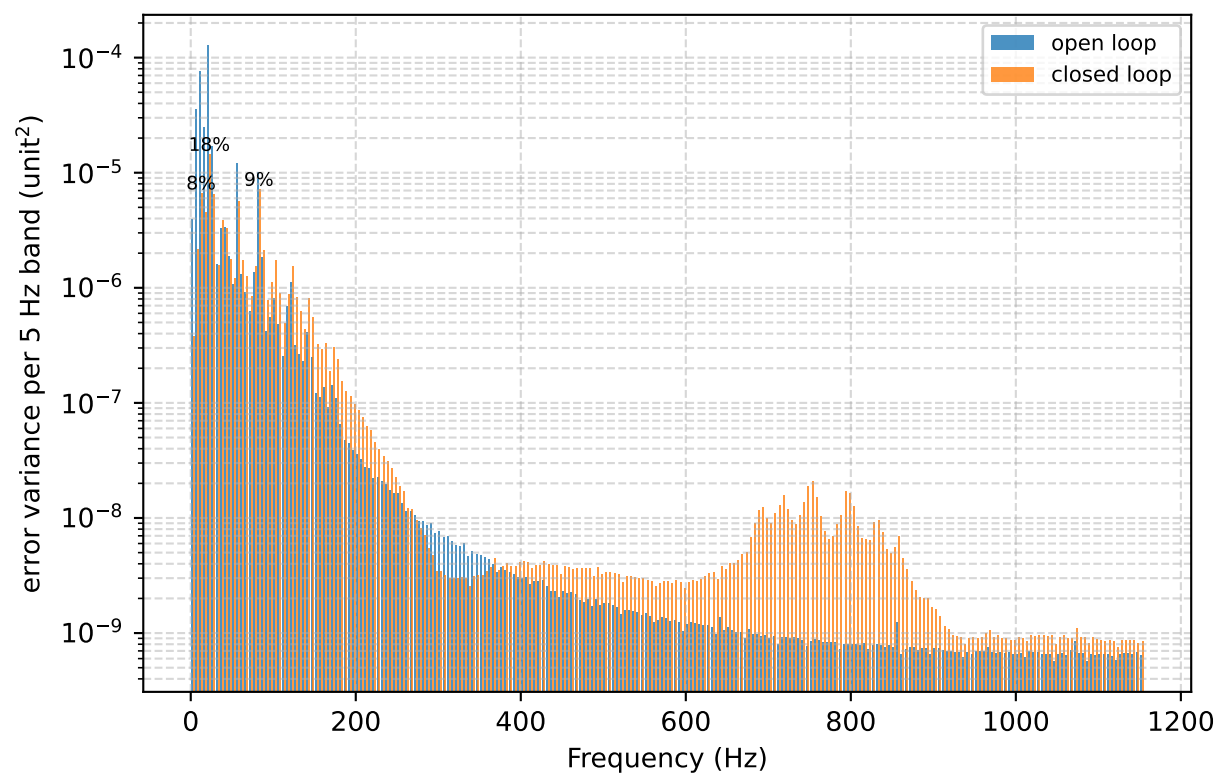
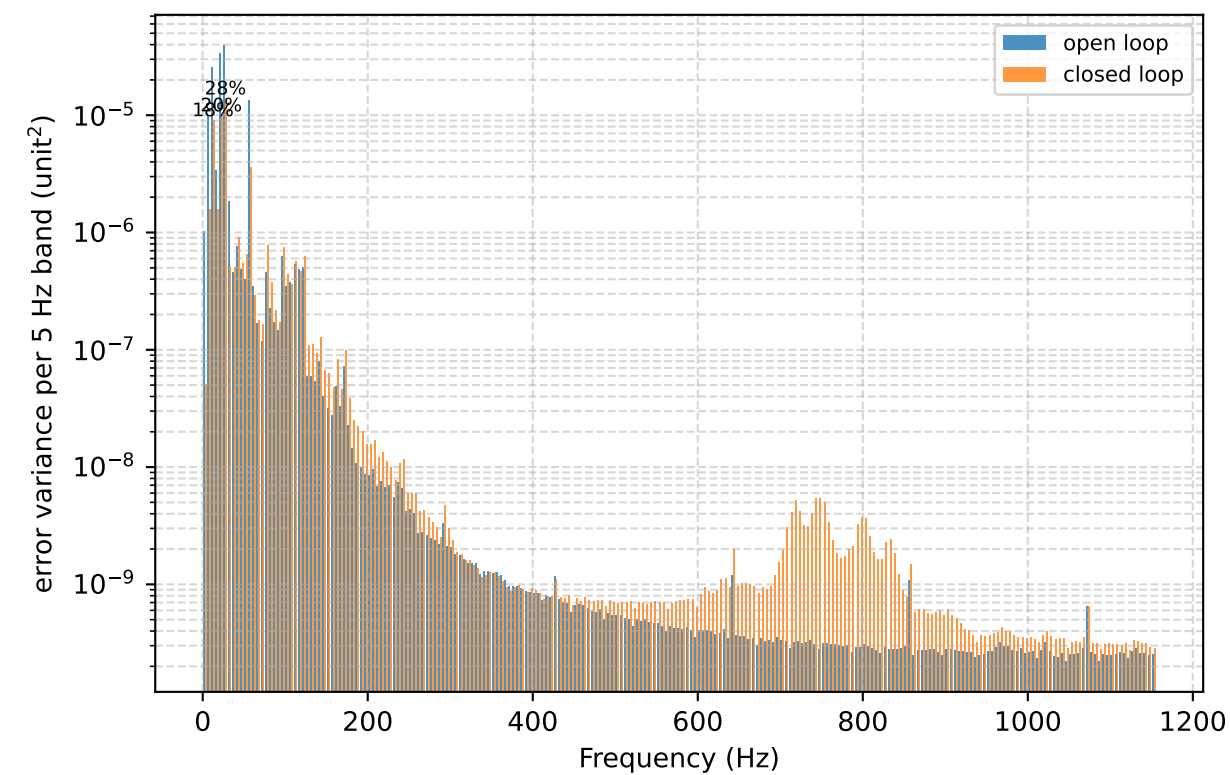
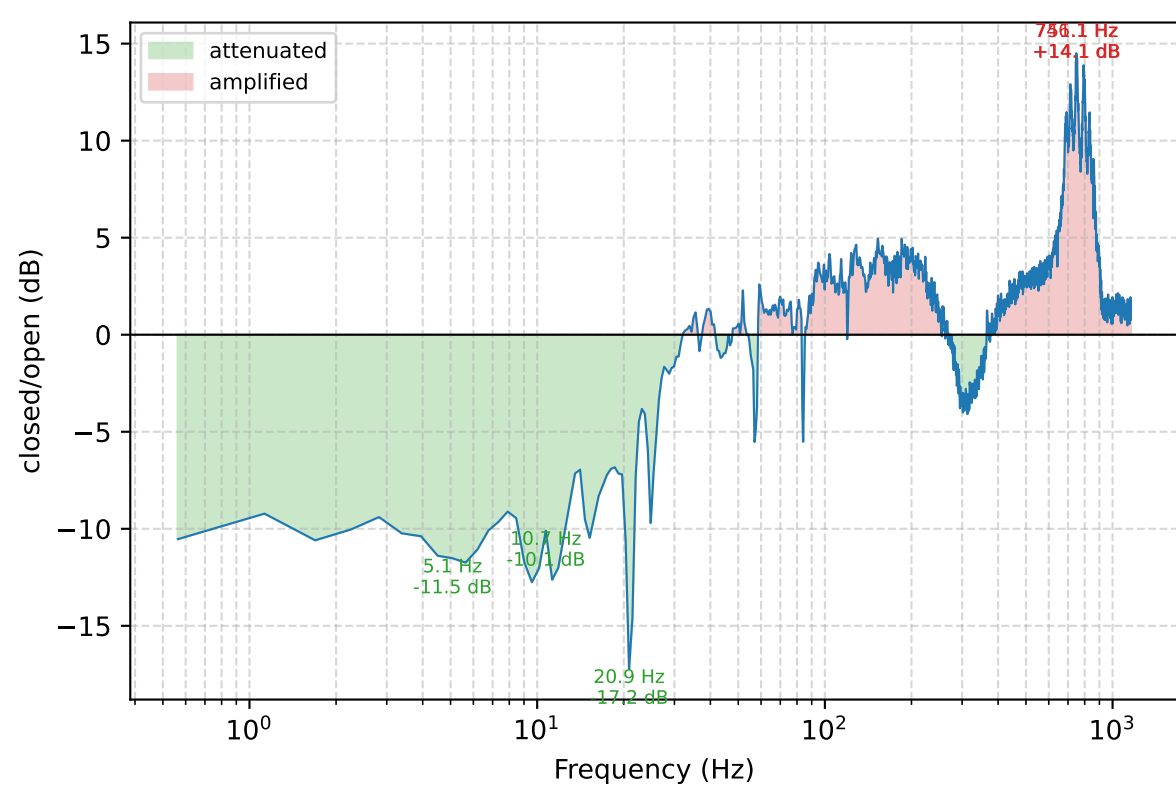
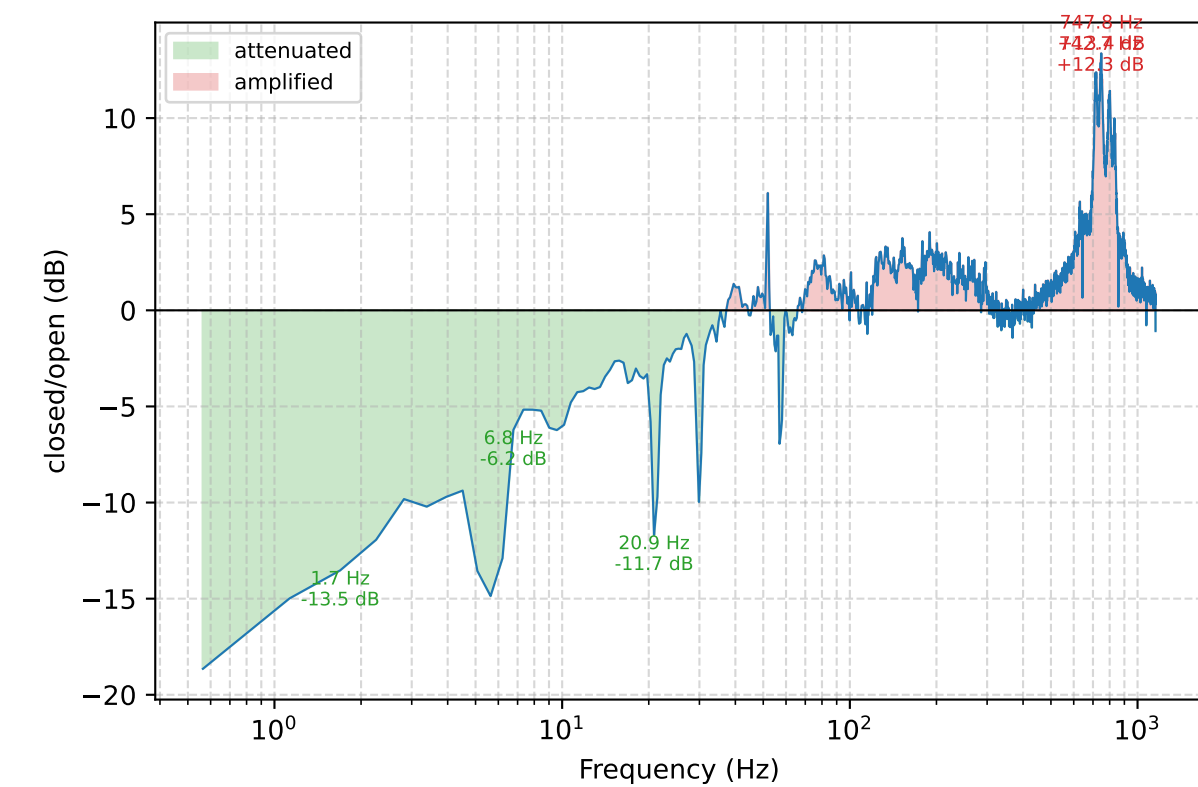
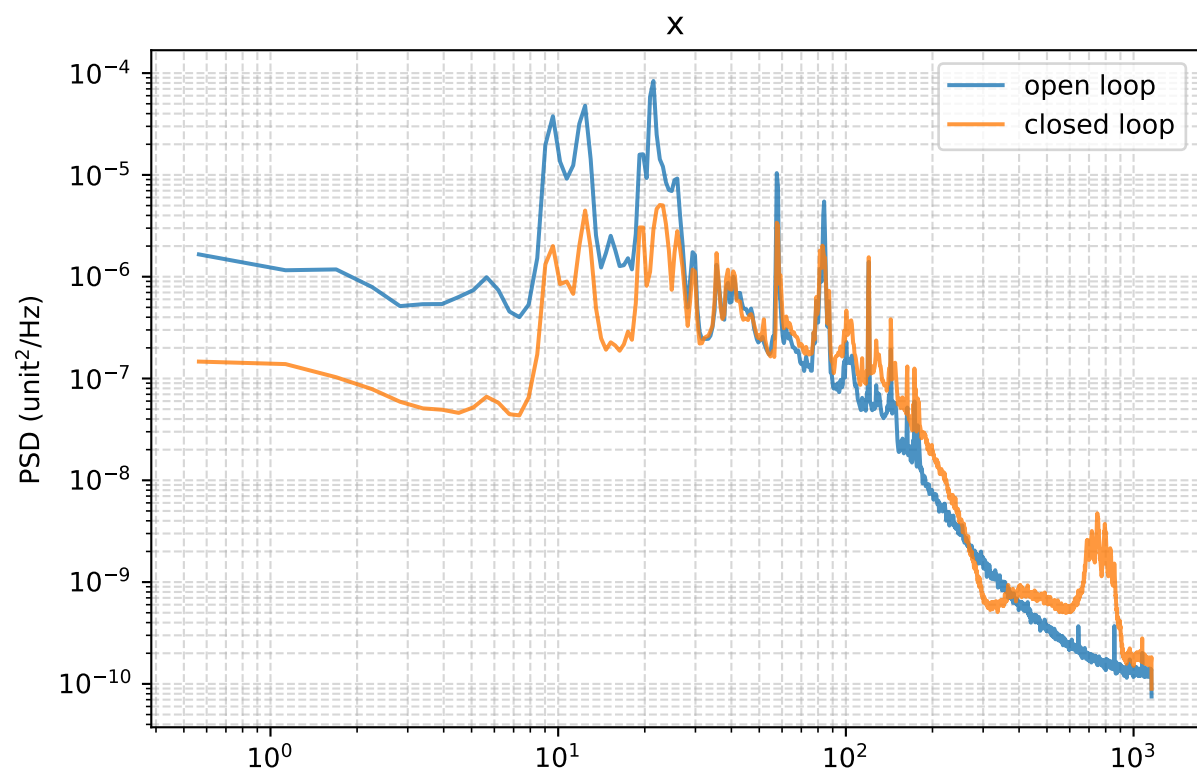
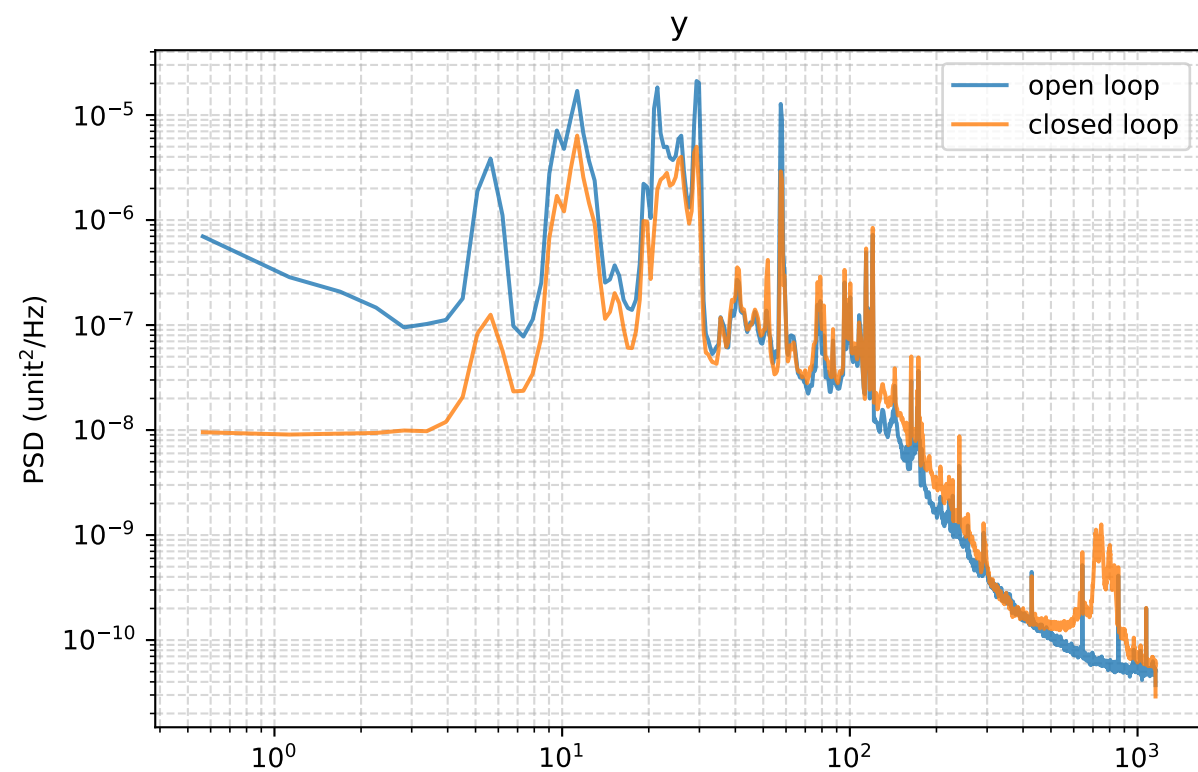


Closed-loop attenuation



Run configuration and per-band RMS

Started: 2026-07-14 11:24:14
pixel_scale: 31.5 px per output unit
y: mean +3.955 -> +0.000 px; RMS about mean 0.461 -> 0.221 px (open -> closed)
x: mean -6.511 -> +0.000 px; RMS about mean 0.662 -> 0.282 px (open -> closed)

Config used:
100, 500, 100, 500, 4096: camera ASI290MM 64x64, ROI start (920,488), ~2310 Hz loop
pid: integrator-only (p=py=d=0), i=680 iy=500, units=minmax (fine bode 2026-07-14 @ bias 2.52/1.62: Kx=0.427 Ky=0.300, tau_loop=1.81 ms; fc_x=46 Hz PM 60, fc_y=24 Hz PM 75)
lines x=[9.8,11.6,21.1,24.9,57.5,83.7] gain=[40,60,60,40,20,15] phase=[77.8,75.6,63.6,58.7,13.8,-24.3]
linesy=[5.64,21.1,29.6,57.5] gainy=[50,30,20,15] phasey=[76.5,42.6,26.8,-13.7]
phases = plant + SENSITIVITY: phi=-360*f*tau + angle(1/(1+L)), from attenuation8 notch-offset analysis
kalman: order=80 horizon=5 mu=0.03 gain=1.0 gainy=1.0 (y predictor ON, first try at fc_y 24 Hz) ramp=10
piplate: fine ch0(+2)/ch1(-2), coarse bias 2.55/1.65 (bode @ 2.52/1.62), baud 460800

RMS contributed by each 10 Hz band (px)

band (Hz)	y open	y closed	x open	x closed
0-10	0.1033	0.0402	0.1976	0.0502
10-20	0.1689	0.1029	0.3160	0.1051
20-30	0.2685	0.1541	0.3781	0.1437
30-40	0.0478	0.0317	0.0694	0.0733
40-50	0.0352	0.0380	0.0721	0.0708
50-60	0.1175	0.0647	0.1143	0.0822
60-70	0.0226	0.0216	0.0468	0.0544
70-80	0.0239	0.0305	0.0444	0.0487
80-90	0.0198	0.0242	0.1091	0.0957
90-100	0.0277	0.0301	0.0311	0.0434
100-110	0.0268	0.0281	0.0358	0.0511
110-120	0.0318	0.0320	0.0306	0.0368
120-130	0.0237	0.0271	0.0377	0.0483
130-140	0.0106	0.0142	0.0222	0.0324
140-150	0.0109	0.0139	0.0256	0.0369
150-160	0.0076	0.0105	0.0152	0.0248
160-170	0.0090	0.0113	0.0150	0.0226
170-180	0.0097	0.0117	0.0158	0.0232
180-190	0.0050	0.0069	0.0105	0.0167
190-200	0.0043	0.0059	0.0091	0.0144
200-210	0.0042	0.0057	0.0082	0.0127
210-220	0.0038	0.0050	0.0073	0.0109
220-230	0.0037	0.0046	0.0067	0.0092
230-240	0.0036	0.0043	0.0063	0.0081
240-250	0.0033	0.0042	0.0058	0.0070
250-260	0.0029	0.0034	0.0054	0.0060
260-270	0.0023	0.0029	0.0048	0.0049
270-280	0.0022	0.0026	0.0044	0.0042
280-290	0.0021	0.0023	0.0042	0.0035
290-300	0.0023	0.0028	0.0040	0.0029
total 0-300	0.3657	0.2214	0.5727	0.2814